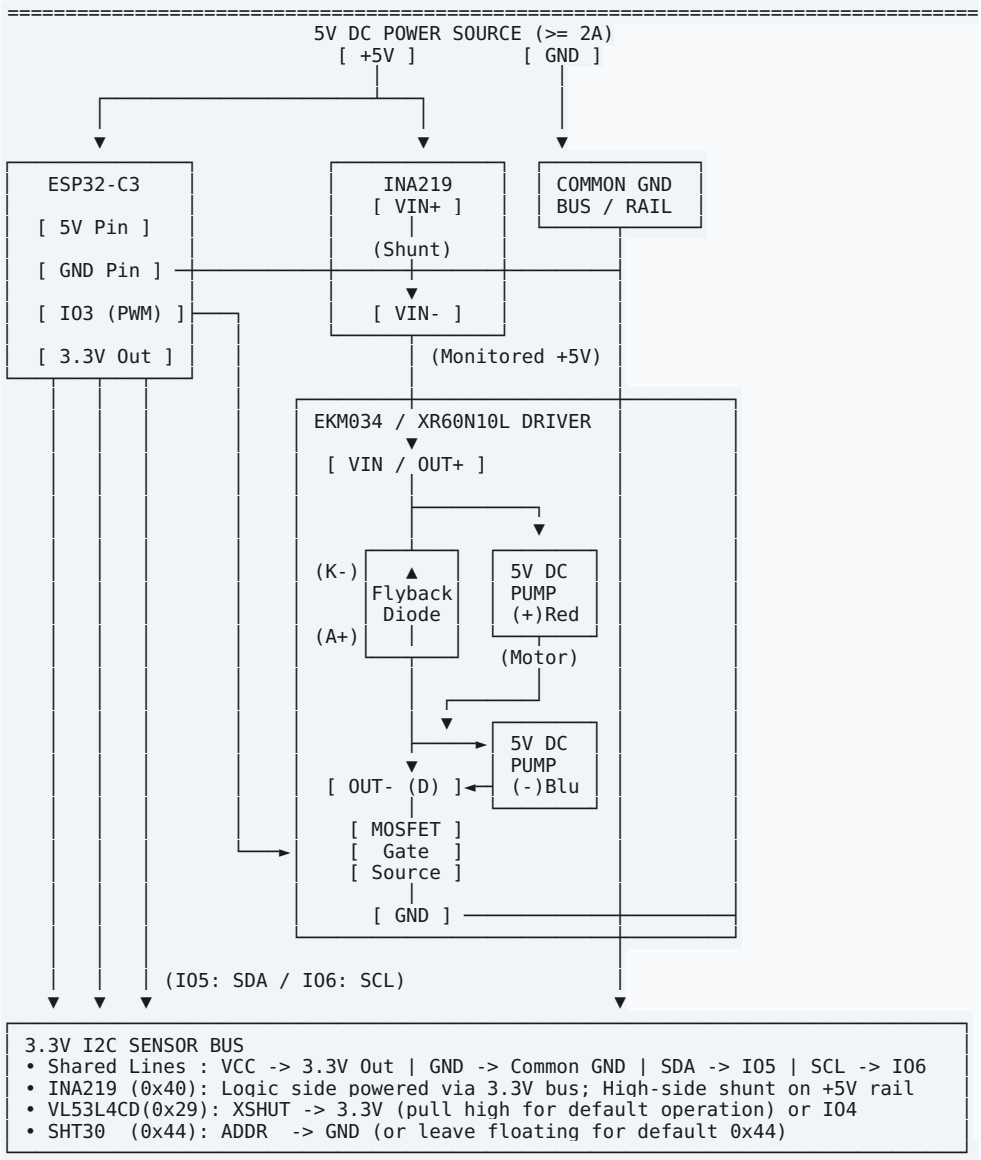
Complete System Wiring Schematic: Water Tank & Pump Controller

This document provides the complete wiring schematic, pin-by-pin hardware connections, and safety checklist for connecting the **ESP32-C3 (0.42” OLED)** with the **VL53L4CD** (distance), **INA219** (high-side current/voltage), **SHT30** (temp/humidity), and **XR60N10L / EKM034** (MOSFET pump driver).

1. Complete System Wiring Schematic



2. Detailed Pin-by-Pin Connections

A. ESP32-C3 0.42” OLED Board Pin Connections

Pin Label on ESP32	Connected To	Wire Role / Notes
5V	5V Power Supply (+)	Main 5V board power
G (GND)	Common Ground Bus	System Ground reference
3V (3.3V)	Sensor 3.3V Bus (VL53L4CD, INA219, SHT30)	Regulated 3.3V logic power
IO5 (D5)	I2C SDA (VL53L4CD, INA219, SHT30)	Shared I2C Data line
IO6 (D6)	I2C SCL (VL53L4CD, INA219, SHT30)	Shared I2C Clock line
IO3 (D3)	EKM034 / XR60N10L module `IN`	Pump Gate On/Off control (Active HIGH)
IO4 (D4)	*Optional:* VL53L4CD `XSHUT`	Hardware reset/enable (or tie XSHUT to 3.3V)

B. High-Power Circuit: INA219, Pump, and XR60N10L (EKM034)

The INA219 measures current on the **High Side** (+5V path) while the XR60N10L switches the **Low Side** (GND return path).

1. INA219 Current Sensor Connections

INA219 Pin	Connects To	Explanation
VCC	ESP32 `3.3V`	**CRITICAL:** Do NOT connect to 5V! Logic must be 3.3V.
GND	Common `GND`	Ground reference for I2C and internal ADC.
SCL	ESP32 `IO6`	I2C Clock
SDA	ESP32 `IO5`	I2C Data
VIN+ (Screw Terminal or Pin)	**+5V Rail** (Power Supply)	Current enters shunt from the 5V supply
VIN- (Screw Terminal or Pin)	EKM034 `VIN / OUT+`	Current leaves shunt and feeds positive driver rail

2. XR60N10L / EKM034 MOSFET Board Connections

Module Pin / Terminal	Connects To	Explanation
IN (Header Pin)	ESP32 `IO3`	Logic signal from MCU: `HIGH` (3.3V) = Pump ON, `LOW` = Pump OFF
GND (Header Pin)	Common `GND`	Logic ground reference for gate signal
VL (Header Pin)	ESP32 `3.3V` (Optional)	Powers onboard gate status LED (can be left open)
VIN / OUT+ (Screw Terminal)	INA219 `VIN-` & Pump (+) Wire	Positive power to load and flyback diode cathode
OUT- (Screw Terminal)	**Pump (-) Wire** (Black/Blue)	Switched low side (MOSFET drain & diode anode)
GND (Screw Terminal)	Common `GND`	High-current power return back to 5V PSU

[!IMPORTANT] **Flyback Diode Integration:** The EKM034 board includes an onboard flyback diode between `OUT-` (anode) and `VIN / OUT+` (cathode). Connecting INA219 `VIN-` and the Pump (+) wire to `VIN / OUT+` clamps the diode across the pump motor, protecting against inductive turn-off spikes.

C. Sensors Bus: VL53L4CD & SHT30

1. VL53L4CD (ToF Distance Sensor)

VL53L4CD Pin	Connects To	Notes
VIN / VCC	ESP32 `3.3V`	3.3V Power
GND	Common `GND`	Ground
SCL	ESP32 `IO6`	I2C SCL (Address `0x29`)
SDA	ESP32 `IO5`	I2C SDA
XSHUT	ESP32 `3.3V` (or `IO4`)	Must be pulled HIGH to enable the sensor
GPIO1 / INT	*Not Connected (NC)*	Optional interrupt (leave floating)

2. SHT30 (Temperature & Humidity Sensor)

SHT30 Pin	Connects To	Notes
VIN / VCC	ESP32 `3.3V`	3.3V Power
GND	Common `GND`	Ground
SCL	ESP32 `IO6`	I2C SCL (Address `0x44`)
SDA	ESP32 `IO5`	I2C SDA
ADDR	Common `GND` (or open)	Address select: Low = `0x44`, High = `0x45`
ALERT / ALR	*Not Connected (NC)*	Hardware threshold alert (leave floating)

3. Hardware Checklist Before Powering On

1. **Power Supply Capacity**: The 5V supply must supply at least **2.0A to 2.5A** to ensure the pump’s inrush current (~1.5A) does not cause the ESP32-C3 to brown out.
2. **INA219 VCC = 3.3V**: Do not connect INA219 VCC to 5V; its I2C lines would pull up towards 5V and risk damaging the ESP32-C3 GPIOs.
3. **Common Ground**: Ensure the ESP32 GND, EKM034 GND, Power Supply (-), and all sensor GND pins are bonded to the same ground rail.